

Exercises 2.1

1) List the members of these sets

a) $\{x | x \text{ is a real number such that } x^2 = 1\}$

Solution: $x \in \mathbb{R}, x^2 = 1 \Rightarrow (x-1)(x+1) = 0 \Rightarrow x = -1 \text{ or } x = 1$

$\therefore$ The solution set: $\{-1, 1\}$.

b) $\{x | x \text{ is a positive integer less than } 12\}$.

Solution: By definition, a positive integer must be greater than 0.

$\therefore$ The solution set: $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$

c) $\{x | x \text{ is the square of an integer and } x < 100\}$.

Solution: The solution set consists of all the perfect squares less than 100.

$\{0, 1, 4, 9, 16, 25, 36, 49, 64, 81\}$

d) $\{x | x \text{ is an integer such that } x^2 = 2\}$.

Solution: $x^2 = 2 \Rightarrow x = \pm\sqrt{2}$. x is not an integer. $\therefore$ The solution set is an empty set. $\emptyset$ (Φ).

2) Use set builder notation to give a description of each of these sets

a) $\{0, 3, 6, 9, 12\}$

Ans: $\{3x | x \in \mathbb{N}, 0 \leq x \leq 4\}$

b) $\{-3, -2, -1, 0, 1, 2, 3\}$

Ans: $\{x | x \in \mathbb{Z}, -3 \leq x \leq 3\}$

c) $\{m, n, o, p\}$

Ans: $\{c | c \text{ is a alphabet in between } l \text{ and } q, \text{ excluding them}\}$

7) Determine whether each of these pairs of sets are equal

a) $\{1, 3, 3, 3, 5, 5, 5, 5, 5\}, \{5, 3, 1\}$.

Soln: Both the sets contain exactly the elements 1, 3, 5. By Defn 2, the sets are equal.

b) $\{\{1\}\}, \{1, \{1\}\}$

Soln: Both the sets contain the element $\{1\}$. But the element 1 is only contained in the second set.